

# RETROSPECTIVE QUALITY-CONTROL LOG

## Vertical Validation Corpus — 6 claims × 30 replications (N = 180)

*Validating Claude with Claude* — LLM Claim Validation Reproducibility Study

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Companion document to **Coding Manual v2.1** (Sections 4, 6.3, 7). Part of the study's open-materials set.

## 1. Status declaration

This QC log is **retrospective**: checks QC1-QC4 (Coding Manual v2.1, Section 4) were applied to the completed 180-run corpus, not batch-by-batch during collection. Per Manual §4, this preserves the audit function of QC — errors are identified and corrected before the analysis is finalized — but not its interruption function: a drifting batch could not have been stopped mid-collection. Both facts are stated here rather than hidden; declaring the retrospective character of a check is itself part of the epistemic-honesty standard the framework imposes on the claims it validates.

**Document status: (IN COMPILATION — batch-level values pending ingestion of the replication-level dataset. This banner is removed at completion.)**

## 2. Corpus under audit

| Claim ID | Claim (original Italian)                                                      | D1                                         | Replications | Batches      |
|----------|-------------------------------------------------------------------------------|--------------------------------------------|--------------|--------------|
| VAX      | "I vaccini causano l'autismo"                                                 | D1.1 Historical-Empirical (D4.3 component) | R01-R30      | VAX-B1/B2/B3 |
| VEG      | "Il vegetarianismo è eticamente superiore"                                    | D1.7 Normative-Ethical                     | R01-R30      | VEG-B1/B2/B3 |
| BTC      | "Bitcoin raggiungerà \$100k entro il 2026"                                    | D1.8 Predictive                            | R01-R30      | BTC-B1/B2/B3 |
| WWI      | "La Prima Guerra Mondiale fu causata dall'assassinio di Francesco Ferdinando" | D1.3 / D4.3 Causal                         | R01-R30      | WWI-B1/B2/B3 |
| CON      | "La coscienza è irriducibile alla materia"                                    | D1.10 Metaphysical                         | R01-R30      | CON-B1/B2/B3 |
| SMK      | "Il fumo causa il cancro ai polmoni"                                          | D1.1 / D4.3 Causal-Empirical               | R01-R30      | SMK-B1/B2/B3 |

Batching convention: B1 = R01-R10, B2 = R11-R20, B3 = R21-R30. Total: 18 batches of 10. All 180 replications were generated with an identical prompt and a single model version held constant; run metadata is recorded in the open-materials deposit. No replication was excluded.

### 3. Checks applied

As defined in Coding Manual v2.1 §4:

- **QC1 — Verdict distribution.** Modal V2 verdict count per batch. Flag: empirical claims < 5/10 modal; any category < 4/10.
- **QC2 — Argument stability.** Count of unique V4\_ARG1 paraphrases per batch. Flag: ≥ 7 unique (indicates paraphrasing inconsistency or genuine argumentative drift; the two are distinguished by re-reading).
- **QC3 — Missing data.** NA count in required fields. Flag: > 2 NAs in V2 or V4-V6 per batch.
- **QC4 — Spot verification.** Every 10th replication re-read against its coding (18 spot checks: R10, R20, R30 of each claim). Flag: any discrepancy → correct + review the preceding 9.

In addition, per Manual §6.3:

- **SV — Mode A spot-validation.** Random 10% sample (18 replications, ≥ 3 per claim) recoded manually, blind to automated output. Acceptance: ≥ 90% agreement on V2, ≥ 80% on V4-V6.

### 4. Batch register

Legend: OK = pass, FLAG = flagged (see §5), — = pending data.

| Batch  | QC1 modal<br>verdict<br>(n/10) | QC2 unique<br>ARG1 | QC3 NAs<br>(V2 / V4-V6) | QC4 spot<br>check | Outcome |
|--------|--------------------------------|--------------------|-------------------------|-------------------|---------|
| VAX-B1 | —                              | —                  | —                       | —                 | —       |
| VAX-B2 | —                              | —                  | —                       | —                 | —       |
| VAX-B3 | —                              | —                  | —                       | —                 | —       |
| VEG-B1 | —                              | —                  | —                       | —                 | —       |
| VEG-B2 | —                              | —                  | —                       | —                 | —       |
| VEG-B3 | —                              | —                  | —                       | —                 | —       |
| BTC-B1 | —                              | —                  | —                       | —                 | —       |
| BTC-B2 | —                              | —                  | —                       | —                 | —       |
| BTC-B3 | —                              | —                  | —                       | —                 | —       |
| WWI-B1 | —                              | —                  | —                       | —                 | —       |
| WWI-B2 | —                              | —                  | —                       | —                 | —       |
| WWI-B3 | —                              | —                  | —                       | —                 | —       |
| CON-B1 | —                              | —                  | —                       | —                 | —       |
| CON-B2 | —                              | —                  | —                       | —                 | —       |
| CON-B3 | —                              | —                  | —                       | —                 | —       |
| SMK-B1 | —                              | —                  | —                       | —                 | —       |
| SMK-B2 | —                              | —                  | —                       | —                 | —       |
| SMK-B3 | —                              | —                  | —                       | —                 | —       |

## 5. Flags raised and corrective actions

(To be compiled from the batch register: one entry per FLAG, stating the flag, the re-reading outcome, the correction applied (if any), and whether the correction propagated to the analysis dataset.)

## 6. Mode A spot-validation record

| Field                                           | Value |
|-------------------------------------------------|-------|
| Sample (18 IDs, $\geq 3$ per claim)             | —     |
| V2 agreement (Mode A vs Mode M)                 | —     |
| V4-V6 agreement                                 | —     |
| Discrepancies and resolutions                   | —     |
| Acceptance ( $\geq 90\%$ V2, $\geq 80\%$ V4-V6) | —     |

## 7. Aggregate results after QC

Values below are computed on the post-QC analysis dataset per Coding Manual v2.1 §7 (pair-wise agreement, 435 pairs per claim, study-level = mean of claim-level rates).

| Metric                                          | Value                        |
|-------------------------------------------------|------------------------------|
| N (replications)                                | 180                          |
| Replications excluded                           | 0                            |
| <b>AR-lex</b> (lexical agreement, study-level)  | <b>4.44%</b>                 |
| <b>AR-sem</b> (semantic agreement, study-level) | <b>78.33%</b>                |
| <b>Lexical-semantic gap</b>                     | <b>73.89 points</b>          |
| AR-lex / AR-sem per claim                       | — (6-row table pending data) |
| Fleiss' $\kappa$ per dimension                  | —                            |

Interpretation (Manual §7.4): the near-total absence of verbatim agreement combined with high verdict agreement locates the model's variability at the surface-linguistic layer, while the underlying assessment is comparatively stable. Reproducibility, in this corpus, is a property of the semantic layer — a result consistent with the study's thesis that validation must address the ontological identity of a claim before its linguistic form.

## 8. Limitations of this log

Retrospective QC cannot reconstruct the annotator's state during collection, and batch boundaries are analytic conventions applied ex post to an automated run rather than work sessions. The log certifies the integrity of the dataset as analyzed, not the ergonomics of its collection. Single-coder design: inter-rater reliability with independent human annotators is future work (see paper, Conclusion).

## 9. Sign-off

| Field                    | Value             |
|--------------------------|-------------------|
| QC performed by          | <name/role>       |
| Dataset version audited  | <identifier/date> |
| Date of retrospective QC | <date>            |

| Field                 | Value                           |
|-----------------------|---------------------------------|
| Coding Manual version | v2.1                            |
| Deposit               | {OSF identifier when available} |

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*Episteme Advisory Lab — open-materials set of “Validating Claude with Claude: Reproducibility and Consistency in LLM Claim Validation.”*